Exercise: Naive Bayes Classification of Wine Data

1. Environment

A Python virtual environment was created with `python -m venv .venv`, activated, and the required packages (numpy, pandas, matplotlib, scikit-learn, ipykernel) were installed into it. The notebook was executed with the `.venv` kernel.

2. Data and method

The UCI Wine dataset contains 178 wines from three cultivars, each described by 13 chemical features. The exploratory plots in the notebook show that features such as Flavanoids, and Proline separate the classes well, but the correlation matrix also shows that many features are correlated, which violates the independence assumption of Naive Bayes. The data was split randomly into 70 % training and 30 % test data. A **GaussianNB** classifier from scikit-learn was trained: it estimates the class priors P(y) and, for every class and feature, a Gaussian mean and variance, then predicts the class.

3. Results

On the test set the classifier reached an accuracy of **0.963**; the confusion matrix and per-class precision/recall are shown in Fig. 1. Because a single random split is noisy, the split and training were repeated 100 times, giving a mean test accuracy of **0.973 ± 0.020** (Fig. 2).

```
Training accuracy: 0.984
Test accuracy:     0.963

Confusion matrix:
 [[18  1  0]
 [ 0 19  1]
 [ 0  0 15]]

Classification report:
              precision    recall  f1-score   support

           1       1.00      0.95      0.97        19
           2       0.95      0.95      0.95        20
           3       0.94      1.00      0.97        15

    accuracy                           0.96        54
   macro avg       0.96      0.97      0.96        54
weighted avg       0.96      0.96      0.96        54
```

Fig. 1. Accuracy, confusion matrix and classification report.

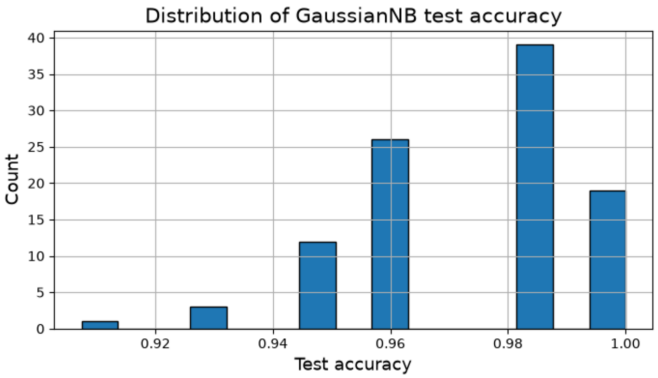


Fig. 2. Test accuracy over 100 random train/test splits.

4. Conclusion

Even though the features are not independent, Gaussian Naive Bayes classifies the wine cultivars very accurately (about 97 % on average) with almost no tuning and very fast training. The few errors occur between cultivars whose feature distributions overlap, mostly involving class 2.